# Methods Review

## RCT, IV and DiD

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Selection Bias

Randomized Controlled Trials

Instrumental Variables

Difference-in-Differences

Putting It All Together

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# Running Example: High-Dosage Tutoring

**Causal question:** What is the (average) causal impact of high-dosage tutoring on 9th-grade math scores?

Setting (stylized numbers): a large urban district offers tutoring in small groups (two students per tutor), one class period every school day. The outcome $Y_i$ is student $i$'s end-of-year math score (SD $\approx 40$ points), and $D_i = 1$ if student $i$ is tutored.

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(Wrong) answer: just compare the average scores of students who chose tutoring against those who didn't...

| Group | Obs | Mean score |
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| No tutoring ($D_i = 0$) | 3,000 | 250 |
| Tutoring ($D_i = 1$) | 1,000 | 246 |
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...and conclude that tutoring *lowers* math scores by 4 points (0.1 SD)?!

# Why is this wrong?

Each student has two *potential outcomes*: $Y_i(1)$, her score with tutoring, and $Y_i(0)$, her score without it. We only observe one of them: $Y_i = D_i Y_i(1) + (1 - D_i)Y_i(0)$.

And students who choose tutoring are different from those who don't:

| | Tutoring | No tutoring | Difference |
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They were already behind: they would probably score lower *even without tutoring*, i.e. they have a lower $Y_i(0)$ on average.

# Selection Bias

Two estimands to start with (a third one, the LATE, comes later):

- $ATE := \mathbb{E}[Y_i(1) - Y_i(0)]$, the average effect for everyone in the population we study.
- $ATT := \mathbb{E}[Y_i(1) - Y_i(0)|D_i = 1]$, the average effect for the treated.

The naive comparison mixes the ATT with *selection bias*:

(No need to follow the math)

$$\underbrace{\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]}_{\text{Naive comparison}} = \underbrace{\mathbb{E}[Y_i(1) - Y_i(0)|D_i = 1]}_{ATT} + \underbrace{\mathbb{E}[Y_i(0)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0]}_{\text{Selection Bias}}$$

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$$-4 \quad = \quad +8 \quad + \quad (-12)$$

In our example (we will see where the +8 comes from very soon): *even without* tutoring, students who opt in would have scored 12 points lower than those who don't (238 vs. 250). The selection bias is large enough to flip the sign!

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# Randomized Controlled Trials

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Intuitively, randomly assigned groups are *on average* identical in everything (observed *and* unobserved), except for the treatment.

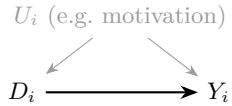
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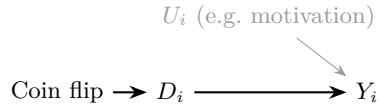
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Intuitively, randomly assigned groups are *on average* identical in everything (observed *and* unobserved), except for the treatment.




Self-selection: $U_i$ drives $D_i$ and $Y_i$




Randomization: nothing else drives $D_i$

# Running Example: A Pilot RCT

The district runs a pilot: 1,000 students volunteer, and a lottery assigns 500 of them to tutoring and 500 to a control group (for now, everyone complies with their assignment).

| | Tutoring | Control | Difference |
|---|---|---|---|
| *Before the lottery (balance check)* | | | |
| 8th-grade math score | 227.1 | 227.4 | −0.3 |
| Low-income (%) | 78 | 77 | +1 |
| English learners (%) | 14 | 13 | +1 |
| *End of the year (outcome)* | | | |
| 9th-grade math score | 246.0 | 238.0 | +8.0 |

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As randomization promises, the groups look alike before the lottery, and the difference in means estimates the ATE among volunteers: about 8 points (0.2 SD).

Bonus: volunteers look like the students who opt into tutoring (both averaged 227 in 8th grade), so the control mean (238) approximates the counterfactual we were missing before: selection bias $\approx 238 - 250 = -12$.

# What Can Go Wrong in an RCT?

RCTs are the "gold standard" of causal inference, but:

- **Imperfect compliance**: some students assigned to tutoring never show up, and some controls find tutoring elsewhere. (IV can help: next!)
- **Attrition**: if students drop out of the data differently across groups (e.g. struggling control students skip the test), the groups stop being comparable.
- **Spillovers**: tutored students may help their classmates, so the control group is partly treated (*interference*).
- **External validity**: effects for 1,000 volunteers in one district may not generalize to other students or places.
- **Feasibility**: randomizing can be too costly, slow, or unethical. (DiD can help here.)

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# Scaling Up: A Tutoring Lottery

Next year, 2,000 students apply for 1,000 seats, so seats are allocated by lottery. Let $Z_i = 1$ if student $i$ wins a seat, and $D_i = 1$ if she actually receives tutoring.

Compliance is now imperfect: some winners never attend (e.g. schedule conflicts), and some losers find similar tutoring elsewhere (e.g. through a nonprofit).

| | Won ($Z_i = 1$) | Lost ($Z_i = 0$) | Difference |
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| Received tutoring (%) | 70 | 10 | +60 |
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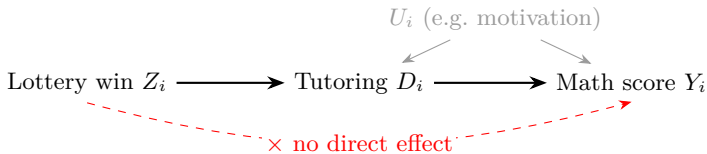
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Comparing tutored vs. non-tutored students brings selection bias back (attending is a choice again)... but winning is still random! Can we use $Z_i$ to learn the effect of $D_i$?

# Instrumental Variables

An **instrumental variable** (or simply an **instrument**) $Z_i$ shifts the treatment $D_i$, but does not *directly* affect the outcome $Y_i$.




Our identification strategy relies on the following assumptions:

1. **Relevance**: $\mathbb{P}[D_i = 1|Z_i = 1] \neq \mathbb{P}[D_i = 1|Z_i = 0]$ (testable: 70% vs. 10%).
2. **Exclusion**: $Z_i$ affects $Y_i$ only through $D_i$ (winning alone does not raise scores).
3. **Monotonicity**: there are no defiers (nobody attends tutoring *only* if she loses).
4. **Independence**: $Z_i$ is independent of potential outcomes and potential take-up (true in a lottery).

# The IV-Wald Estimator

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3. Divide the difference in outcomes by the difference in exposure:

$$\text{IV-Wald} \rightarrow \frac{\mathbb{E}[Y_i|Z_i = 1] - \mathbb{E}[Y_i|Z_i = 0]}{\mathbb{E}[D_i|Z_i = 1] - \mathbb{E}[D_i|Z_i = 0]} = \frac{ITT}{\text{First Stage}} = \frac{6}{0.60} = 10$$

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$\therefore$ tutoring raises math scores by 10 points (0.25 SD)... but *for whom*?

# Local Average Treatment Effect (LATE)

Our target estimand is the **Local Average Treatment Effect** (LATE), the average effect on the *compliers*:

$$LATE := \mathbb{E}[Y_i(1) - Y_i(0) | D_i(Z_i = 1) > D_i(Z_i = 0)]$$

A taxonomy of the applicants (we can back out the shares from the lottery table):

- Always-takers (10%): tutored even if they lose (the 10% of losers who got tutoring).
- Never-takers (30%): not tutored even if they win (the 30% of winners who skipped).
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Why is the LATE (10) larger than the ITT (6)? Winning changes nothing for always- and never-takers, so the ITT is *diluted*: $ITT = 0.6 \times 10 + 0.4 \times 0 = 6$.

# In the Real World: Tutoring in Chicago

Guryan et al. (2023, AER) randomized offers of high-dosage math tutoring (50 minutes every school day, two students per tutor) among more than 5,000 students in grades 9 and 10 in Chicago public high schools.

Not every student offered tutoring participated, so they use random assignment $Z_i$ as an instrument for participation $D_i$, just like in our lottery:

- ITT (study 1): 0.09 SD in math
- IV, i.e. the effect of participating for compliers (study 1): 0.18 SD in math

Across their two studies, participating raised math scores by 0.18 to 0.40 SD, and the effects persisted into future years.

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# No Lottery? A Policy Rollout

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| Average math score | 2024 ($Pre$) | 2025 ($Post$) | $Post - Pre$ |
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| North − South | −16 | −8 | +8 |

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- Post-period comparison: $242 - 250 = -8$ (selection bias: North was already behind).
- Before-after comparison in North: $+10$ (mixes the effect with the district-wide trend).
- **Difference-in-differences**: $(+10) - (+2) = +8$.

# Difference-in-Differences

Let $D_i = 1$ for students in North (the treatment group). Our target estimand is the ATT in the post-treatment period:

$$ATT := \mathbb{E}[Y_i^{Post}(1) - Y_i^{Post}(0) | D_i = 1]$$

The DiD estimator only needs 4 sample averages (1 = treatment group, 0 = control group):

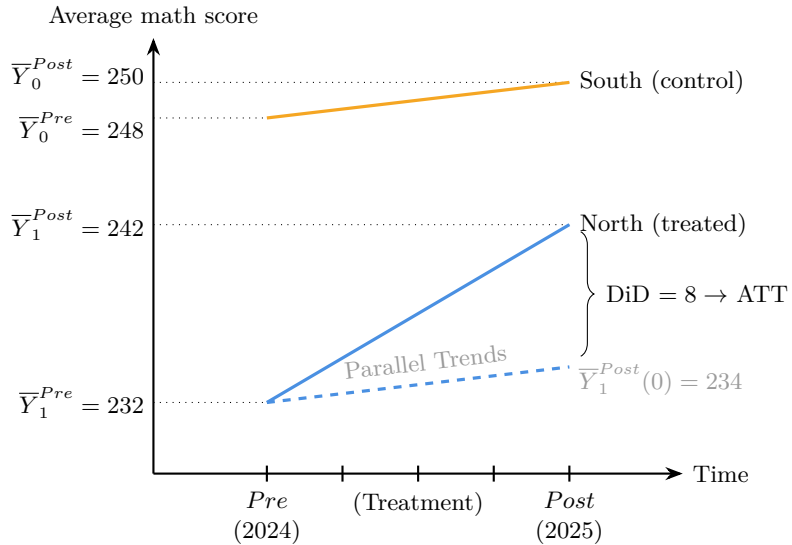
$$(\overline{Y}_1^{Post} - \overline{Y}_1^{Pre}) - (\overline{Y}_0^{Post} - \overline{Y}_0^{Pre}) = (242 - 232) - (250 - 248) = 8$$

It estimates the ATT if the **parallel trends** (PT) assumption holds:

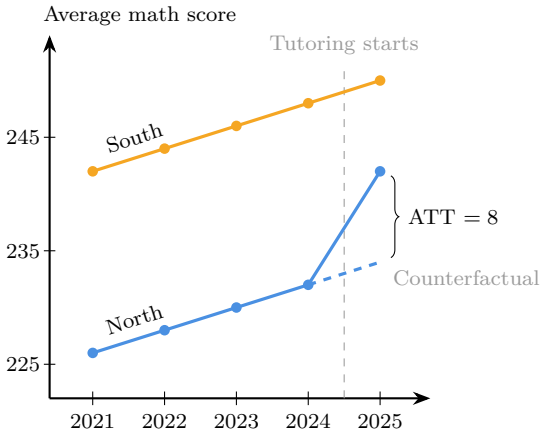
$$\mathbb{E}[Y_i^{Post}(0) - Y_i^{Pre}(0) | D_i = 1] = \mathbb{E}[Y_i^{Post}(0) - Y_i^{Pre}(0) | D_i = 0]$$

In words: had North not adopted tutoring, its scores would have grown as much as South's did (+2).

# DiD Visualized



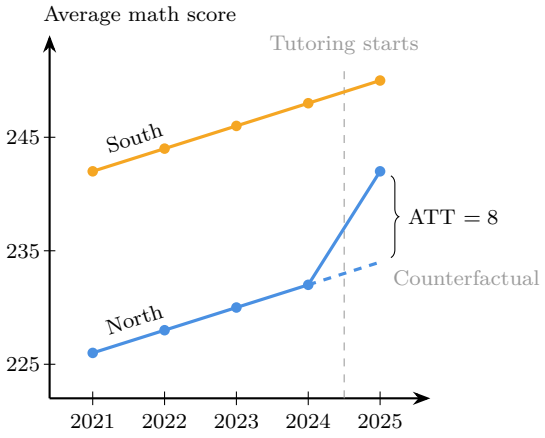
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Threats to PT:

- Something else changed in North in 2025 (e.g. a new curriculum).
- Mean reversion: North was picked right after an unusually bad year.
- Composition: families move in or out in response to the policy.

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# One Question, Four Answers

| | Naive comparison | RCT | IV | DiD |
|---|---|---|---|---|
| Compares | Takers vs. non-takers | Treatment vs. control | Winners vs. losers, scaled by take-up | Changes in North vs. South |
| Estimand | ATT + selection bias | ATE (volunteers) | LATE (compliers) | ATT |
| Key assumption | No selection bias (!) | Random assignment | Relevance, exclusion, monotonicity, independence | Parallel trends |
| Our estimate | $-4$ | $+8$ | $+10$ | $+8$ |
| In the literature | | Project STAR (Krueger, 1999) | School spending (Jackson et al., 2016) | Teacher bargaining (Lovenheim & Willén, 2019) |

# Takeaways

- A naive comparison = causal effect + selection bias (which can even flip the sign).
- **RCTs** eliminate selection bias by design: a difference in means estimates the ATE for the students in the experiment.
- **IV** uses a random "nudge": $LATE = ITT/\text{First Stage}$, the effect on compliers.
- **DiD** compares changes instead of levels: it estimates the ATT if trends would have been parallel.
- Each design targets its own estimand under its own assumptions, so their estimates need not coincide.

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- Each design targets its own estimand under its own assumptions, so their estimates need not coincide.

**Question:** The IV estimate (+10) is larger than the RCT estimate (+8). Does that mean one of them is wrong?

# Backup: Why Does the IV-Wald Estimator Recover the LATE?

(No need to follow the math)

Write $D_i(z)$ for student $i$'s treatment status if $Z_i = z$ (i.e. $D_i(Z_i = z)$). Compliers have $D_i(1) = 1$ and $D_i(0) = 0$.

$$\begin{aligned}
\mathbb{E}[Y_i|Z_i = 1] - \mathbb{E}[Y_i|Z_i = 0] &= \mathbb{E}[Y_i(D_i(1)) - Y_i(D_i(0))] && \text{(Indep. + Exclusion)} \\
&= \mathbb{E}[(Y_i(1) - Y_i(0))(D_i(1) - D_i(0))] \\
&= \mathbb{P}[\text{complier}] \cdot \mathbb{E}[Y_i(1) - Y_i(0)|\text{complier}] && \text{(Monotonicity)}
\end{aligned}$$

$$\mathbb{E}[D_i|Z_i = 1] - \mathbb{E}[D_i|Z_i = 0] = \mathbb{E}[D_i(1) - D_i(0)] = \mathbb{P}[\text{complier}] \qquad \text{(Indep. + Monotonicity)}$$

Dividing the first expression by the second: IV-Wald $\rightarrow$ LATE. (Relevance guarantees that we are not dividing by zero.)

# Backup: DiD as a Linear Regression

With student-level data from both years, a regression gives us the same number:

$$Y_{it} = \alpha + \beta Treat_i + \gamma Post_t + \delta\, Treat_i \times Post_t + \varepsilon_{it}$$

Each coefficient maps to our 4 averages:

$$\begin{aligned}
\mathbb{E}[Y_{it}|Treat_i = 0, Post_t = 0] &= \alpha && = 248 \\
\mathbb{E}[Y_{it}|Treat_i = 1, Post_t = 0] &= \alpha + \beta && = 232 && \Rightarrow \beta = -16 \\
\mathbb{E}[Y_{it}|Treat_i = 0, Post_t = 1] &= \alpha + \gamma && = 250 && \Rightarrow \gamma = +2 \\
\mathbb{E}[Y_{it}|Treat_i = 1, Post_t = 1] &= \alpha + \beta + \gamma + \delta && = 242 && \Rightarrow \delta = +8
\end{aligned}$$

So $\hat{\delta}$ is exactly the DiD estimate. Regressions make it easy to add more groups and years (fixed effects), but when groups adopt the policy at different times and effects vary, this simple regression can mislead (newer DiD estimators fix this).